SMART INDIA HACKATHON 2025

CAREER MENTOR

- **Problem Statement ID –** SIH25094
- **Problem Statement Title-** One-Stop

Personalized Career & Education Advisor

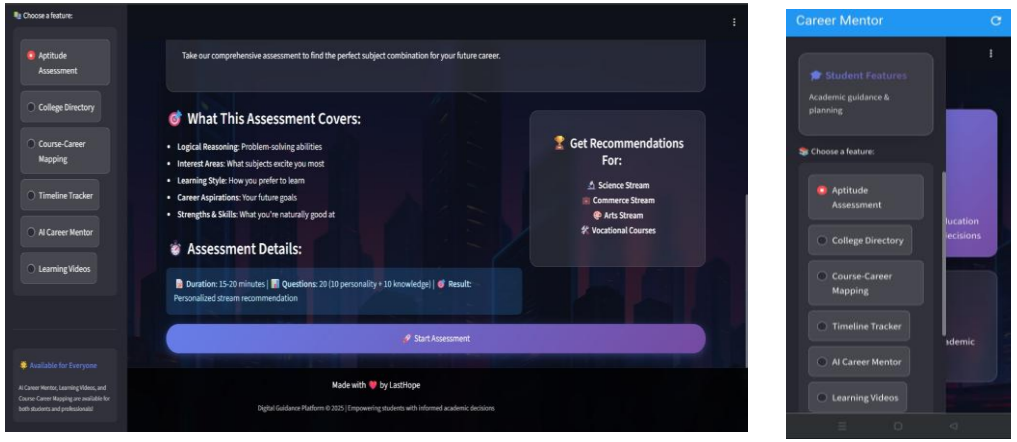
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-** 55038
- **Team Name (Registered on portal)-** LastHope

CAREER MENTOR

❖ Proposed Solution

CareerMentor acts like a **one-stop smart guide** that gives students the **missing information** and **personalized advice** they need, so they can confidently **choose** the right stream, college, and career. Our solution addresses the core problem through a multi-faceted approach.

- Our **Aptitude Assessment** page helps students discover their ideal academic stream through a mix of personality, knowledge, and interest-based questions, followed by AI-driven career guidance.
- **Government College Directory**: Location-based search for government colleges with AI-powered detailed info (courses, fees, facilities, placements, scholarships, timelines).
- **Timeline Tracker** acts as a **personal academic calendar**, keeping students updated with admission deadlines, entrance exams, and scholarship notifications through a smart, real-time tracker with custom alerts.
- **Additional Features:**
  - **Job recommendations with real-time market insights, salary data, and skill improvement resources for College Students/Working Professionals** to explore alternative career paths and upskilling opportunities.
  - **Resume Analyzer** that evaluates resumes for clarity, skill relevance, impact, and ATS compatibility, while also giving **personalized improvement suggestions**.



Prototype Preview

❖ Innovation and Uniqueness:

- ✓ Web + Mobile app ensures accessibility across devices
- ✓ Live data feeds from government college databases and job market trends
- ✓ Peer mentoring and alumni connect modules
- ✓ Automated alerts for admission windows and scholarship deadlines
- ✓ Multilingual support for regional accessibility

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TECHNICAL APPROACH

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- **Aptitude Assessment:**
Integrated the **OpenTrivia API** with personality and interest questions, analyzed responses using **Mistral AI** to generate personalized stream recommendations with confidence scores, career paths, and relevant college options.

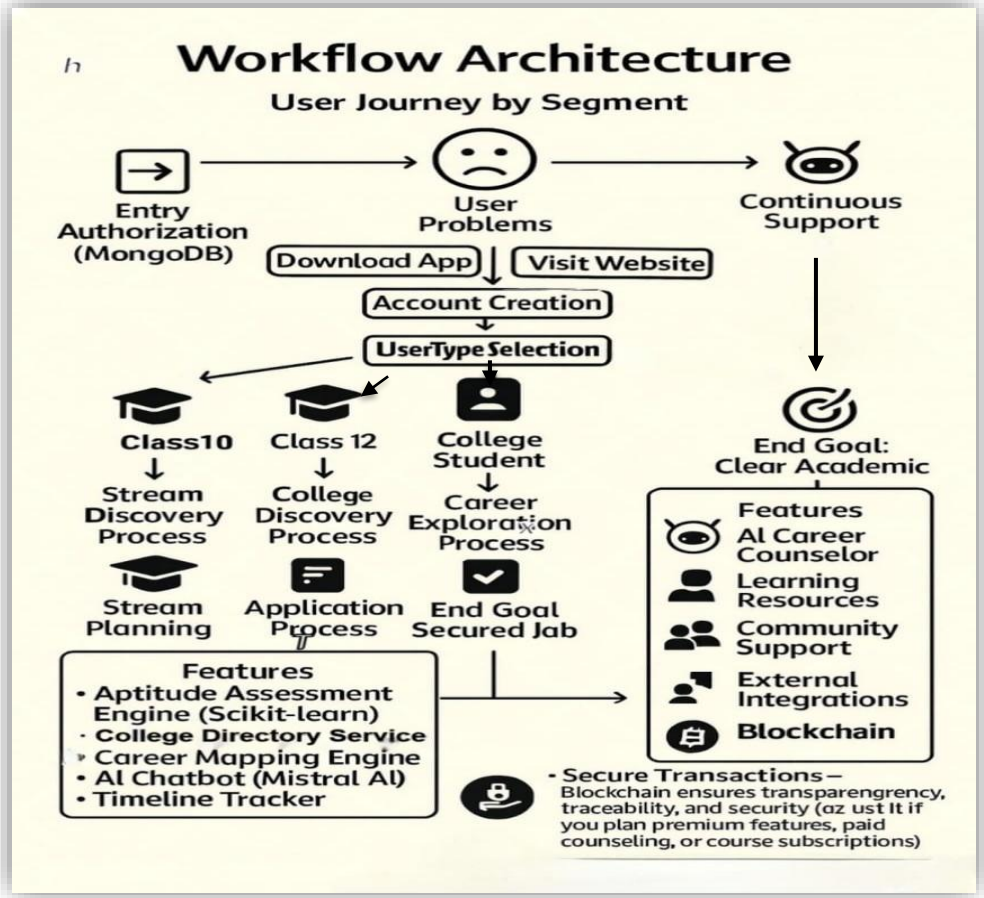
- **Government College Directory:**
This module uses **OpenCage API for geolocation** and **Google Search API** for real-time government college details (courses, fees, facilities, timelines).

- **Timeline Tracker:**
This module leverages **SerpAPI** to fetch real-time admission deadlines, entrance exam schedules, and scholarship updates, offering a **personal academic calendar.**

- **Job Recommendation Engine:**
Our engine combines **Machine Learning models (PyTorch)** with **Google Search API** to provide highly relevant, personalized career paths. Used **Scikit-learn.**

- **Resume Analyzer:**
Evaluates resumes using **PyTorch NLP models** for feature extraction and rating, **PDFPlumber & python-docx** for text extraction, **Google Search API** to fetch real-time best practices and recruiter trends, presents via a **Streamlit** interactive dashboard with feedback.

- **Other Technologies**
Flutter with WebView for cross-platform compatibility
BCrypt for secure password hashing; and
a future **blockchain-based** subscription model.



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FEASIBILITY AND VIABILITY

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Feasibility:

1. Technology: Utilizes existing technologies (web, mobile, AI, API, Cloud)
2. Market: 500+ million students have Internet access in India
3. Economic: Low operational costs due to cloud-based architecture
4. Operational: Continuous content updates through automated systems

Viability:

1. First platform focused specifically on government colleges
2. End-to-end guidance from Class 10 to career
3. 40% dropout rate in government colleges due to poor guidance
4. 85% of students report lack of career awareness

Challenges:

1. Government college information may be outdated
2. Rural areas with poor connectivity
3. Students unfamiliar with technology
4. Traditional counselling preferences

Overcome:

1. Regular automated data validation processes
2. SMS-based alerts for critical updates
3. Video tutorials and guided walkthroughs
4. Incentive programs for early adopters

Business Potential:

1. 15+ million Class 10/12 students annually in India
2. 8+ million college students and young professionals
3. Parents and educators (30+ million stakeholders)
4. Government college focus, AI-powered personalization

Revenue Model:

1. Premium Subscription for advanced features.
2. Advanced AI insights, priority support, detailed reports
3. State education department collaborations
4. Anonymized insights for education policy makers

★ Supporting Facts for Feasibility and Viability ★

- India is projected to have **over 900 million internet users by 2025**, with a significant portion being students and youth. Sourece : Business Standard – IAMAI & Kantar Report
- A staggering **93% of Indian students aged 14–21 are aware of only seven traditional career options**, despite there being over 250 viable paths. Source: IndiaCareerCentre – Career Awareness Gap
- India saw **over 44 lakh students** appear for Class 10 and 12 board exams in 2025, with **23.7 lakh in Class 10** and **17 lakh in Class 12** alone. Source: Indian Higher Education Factsheet – PIB

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IMPACT AND BENEFITS

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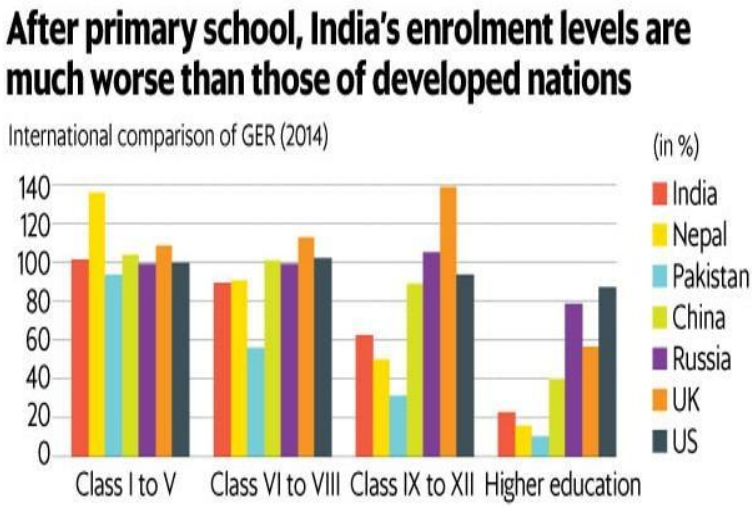

❖ Initial Impact on the target audience:

•**Career Decision**: 40% increase in informed career decisions
•**Course Selection**: Reduced confusion about course selection
•**Interest Alignment**: Better alignment between interests and career choices
•**Government Colleges**: Improved awareness of government college opportunities
•**Student-Course Fit**: Better fit leading to lower dropout rates
•**Infrastructure Utilization**: Better utilization of government education infrastructure
•**Policy Insights**: Data-driven insights for educational policy improvements

❖ Benefits:

Social Benefits:
- Democratized access to career guidance.
- Reduced educational inequality.
- Empowered rural and underprivileged students.
- Strengthened government education ecosystem.

Economic Benefits:
- Better job market alignment.
- Increased skilled workforce in government sector.
- Economic development through education.

Environmental Benefits:
- Reduced paper usage through digital processes.
- Decreased travel for college information gathering.
- Sustainable technology practices.

Educational Benefits:
- Helps students from rural areas discover government colleges.
- Provides timely updates to prevent missed opportunities.
- Supports lifelong learning by linking skills to career pathways.

Long-term Benefits:
- Cultural shift towards valuing government education.
- Improved employment rates among graduates.
- Enhanced social mobility for disadvantaged communities.
- Stronger foundation for India's digital economy.

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RESEARCH AND REFERENCES

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❖ References

1. "Career Guidance and Student Decision Making" - Journal of Educational Psychology
2. "Digital Divide in Rural Education" - Indian Education Review
3. "Government College Enrollment Trends" - Ministry of Education Reports
4. "Assessment of the Digital Divide in Rural Education" - Global International Scientific Research Journal
5. "Career Decision-Making in Students" - Frontiers in Education

Demo: CLICK HERE

❖ Comparison with Existing Systems

Feature	Career_Mentor	Careers360	Shiksha.com	Collegedunia	Our Advantage
Government College Focus	✓ Specialized	✗ Limited	✗ Limited	✗ Limited	Unique positioning
AI-Powered Aptitude Assessment	✓ Advanced ML	✗ Basic	✗ Basic	✗ None	Personalized recommendations
Real-time Admission Updates	✓ SerpAPI	✗ Manual	✗ Manual	✗ Manual	Live data feeds
Visual Career Mapping	✓ Interactive	✗ Text	✗ Text	✗ Limited	Better user experience
Mobile App	✓ Flutter Web	✗ Web-only	✗ Web-only	✓ Basic	Accessibility advantage
AI Chatbot Counselor	✓ Contextual AI	✗ None	✗ None	✗ None	24/7 personalized guidance
Timeline Tracker	✓ Smart Notifs	✗ Basic	✗ Basic	✗ None	Proactive alerts
Free Access	✓ Freemium	✓ Freemium	✓ Freemium	✓ Freemium	Accessibility for all
Multi-language Support	✓ Regional Lang	✗ English	✗ English	✗ Limited	Rural accessibility
Community Features	✓ Peer Support	✗ None	✗ None	✗ Limited	Social learning
Resume Analysis	✓ AI-Powered	✗ None	✗ None	✗ None	Professional development
YouTube Integration	✓ Curated	✗ None	✗ None	✗ None	Rich learning resources